

MOHAMMED ZAKWAN M

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linkedin.com/in/mohammed-zakwan-m | github.com/zakwanmunawar | Portfolio

TECHNICAL SKILLS

Core Programming: Python (Proficient), C, Java, SQL, JavaScript, HTML/CSS (Fundamentals)
Computer Vision & AI/ML: Object Detection (YOLO Architectures v1-v26), From-Scratch Model Implementation, Image Processing, Deep Learning Pipelines, Neural Networks
Frameworks & Libraries: PyTorch, OpenCV, NumPy, Pandas, Matplotlib, Scikit-Learn
Developer Tools: Git, GitHub, VS Code, Google Colab, Jupyter Notebook, Antigravity
Core CS Concepts: Data Structures & Algorithms (DSA), Object-Oriented Programming (OOP), Database Management Systems (DBMS), Operating Systems

EDUCATION

Bachelor of Technology in Computer Engineering (Artificial Intelligence & Machine Learning) **2023 - 2027**
Presidency University, Bangalore, Karnataka **CGPA: 7.24 / 10.0**
Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming, Database Management Systems, Software Engineering, Storage Architecture

PROJECTS

Aerial & UAV Object Detection & Tracking System | *Python, PyTorch, YOLO, OpenCV, TensorRT, ONNX*
github.com/zakwanmunawar/AeroDrop-Tracker

- Trained a 2.37M-parameter YOLO26n object detection model on a 12-class aerial benchmark dataset (86,000+ annotations), achieving 22.1% mAP@50 across all target classes under high background clutter and object density.
- Optimized real-time execution pipelines on an NVIDIA Tesla T4 GPU, achieving sub-3ms total frame latency (330+ FPS raw inference speed).
- Applied dataset preprocessing, frame resizing, and bounding-box normalization to improve detection accuracy across varying background conditions.
- Integrated OpenCV for real-time video feed inference with dynamic bounding-box tracking and minimal latency.

Construction Worker Safety & PPE Compliance Detector | *Python, YOLO, OpenCV*
github.com/zakwanmunawar/worker-safety-system

- Developed and fine-tuned a 3M-parameter YOLO object detection pipeline for workplace site safety, achieving 97.0% mAP@50 and 96.5% precision across 3 critical violation classes (Fall, Helmet, Jacket).
- Optimized video feed inference pipelines on an NVIDIA Tesla T4 GPU, processing multi-class site compliance alerts at sub-9ms frame latency (117+ FPS end-to-end execution).
- Implemented OpenCV video feed processing to overlay dynamic, color-coded bounding boxes flagging missing equipment for automated site monitoring.

CERTIFICATIONS & ACHIEVEMENTS

YOLO Architecture Implementation & Research Series (v1 to v26) - PyTorch & Google Colab
Implemented YOLOv1 from scratch in PyTorch; systematically analyzing and implementing object detection model progressions across YOLO iterations.

Virtual Internship Certification in Python Programming - CodSoft [2025]
Successfully completed a 4-week virtual program developing modular Python scripts, core algorithmic logic, and OOP concepts.

Data Structures & Algorithms in Python - Active Problem Solving on LeetCode
Implemented core algorithms, recursion, and data structures strictly in Python.

Top 70 Selection (out of 500+ Teams) - University Innovation & Project Review (IPR)
Recognized for hardware-software integration on the Autonomous Robotic Car Arm project.